Math Kangaroo Practice

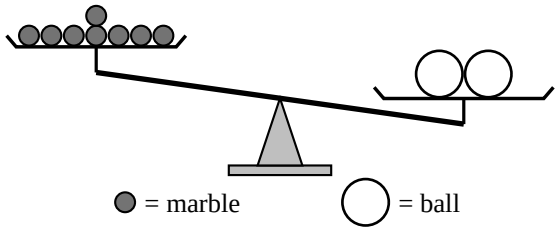
Grade 3-4 · 20 questions · harder · seed 11178

Part A: Each question is worth 3 points

1. Flea the grasshopper's longest jump in training was 50 dm 60 cm and 30 mm long. At the competition the jump was 130 cm longer.
How long was the jump at the competition?

A. 2 m 43 cm ***B.*** 2 m 70 cm ***C.*** 5 m 63 cm ***D.*** 6 m 93 cm ***E.*** 18 m 63 cm

2. On one plate of a balance there are 8 marbles and on the other there are 2 balls. When we put one more ball, exactly like the others, on the plate with the marbles, the balance is level.



One ball weighs the same as how many marbles?

A. 16 marbles ***B.*** 11 marbles ***C.*** 10 marbles ***D.*** 9 marbles ***E.*** 8 marbles

3. Martha has the cards 1, 4 and 8. Martha must use every card exactly once, with +, –, ×, ÷ and brackets.
Which of these numbers can Martha get?

A. 34 ***B.*** 36 ***C.*** 38 ***D.*** 42 ***E.*** 43

4. Nora, Sofia, Chloe, Julien and Aisha were born in different months: July, March, January, December and August. Aisha and Julien both have a birthday in summer. Aisha was born in a month that starts with J. Nora was not born in spring. Sofia was born in a month that starts with J. Who was born in March?

A. Chloe ***B.*** Aisha ***C.*** Sofia
D. Nora ***E.*** Impossible to determine

5. Ben, Iris, Felix and Grace stand in a row for a photo. Each time they stand in a different order. How many different photos can they take?

A. 24 ***B.*** 25 ***C.*** 27 ***D.*** 28 ***E.*** 30

6. Hoppy the frog's longest jump in training was 30 dm 50 cm and 50 mm long. At the competition the jump was 122 cm longer.
How long was the jump at the competition?

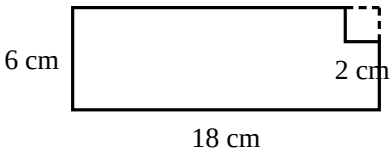
A. 3 m 55 cm ***B.*** 4 m 77 cm ***C.*** 5 m 22 cm ***D.*** 5 m 77 cm ***E.*** 15 m 75 cm

Part B: Each question is worth 4 points

7. A counter shows 9476.
How much must it count up until it shows a number with all 4 digits the same?

A. 634 ***B.*** 588 ***C.*** 571 ***D.*** 524 ***E.*** 523

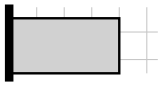
8. A rectangle is 18 cm long and 6 cm wide. A square with side 2 cm is cut off one corner.



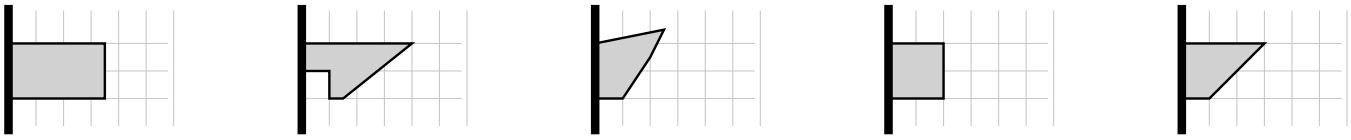
What is the perimeter of the remaining shape?

A. 56 cm ***B.*** 52 cm ***C.*** 48 cm ***D.*** 44 cm ***E.*** 40 cm

9. From a window, Mia sees a rectangular flag on a pole. The wind turns and folds the flag, but the side sewn to the pole never moves. Mia draws the shadow of the flag on the wall at five different moments. Flat, the flag is 4 squares wide and 2 squares tall (see the picture). Some of the drawings are wrong.



the flag, flat

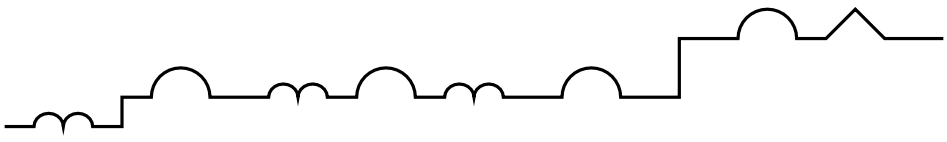


A B C D E

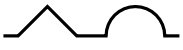
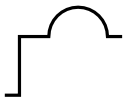
How many of the five drawings can be right?

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

10. Lena cut a strip of paper along this line.



Which piece **can** come from the cut line?

A.  ***B.*** ***C.*** ***D.*** ***E.*** 

11. Mia builds a brick 5 cubes long and 4 cubes tall from red and blue cubes. The outside is completely red and the inside is completely blue. Mia used exactly 6 blue cubes.
How many cubes deep is the brick?

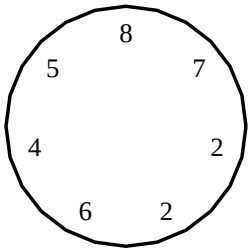
A. 2 ***B.*** 3 ***C.*** 4 ***D.*** 5 ***E.*** 6

12. A counter shows 19710.
How much must it count up until it shows a number with all 5 digits the same?

A. 2512 ***B.*** 2511 ***C.*** 2496 ***D.*** 2036 ***E.*** 1

Part C: Each question is worth 5 points

13. Numbers are written around a round plate as shown. Tom wants to break the plate with one straight line so that the two pieces have the same sum of numbers.



In how many different ways can he do it?

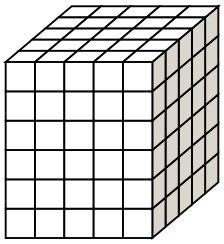
A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

14. Emma, Omar, Nicolas, Noah and Fabienne were born in different months: March, October, June, December and April. Fabienne was not born in spring. Noah has a birthday in summer. Emma was not born in March. Nicolas was born in winter.
Which statement is **not** true?

A. Emma was born in April ***B.*** Noah was born in June ***C.*** Fabienne was born in October

D. Noah was born in March ***E.*** Nicolas was born in December

15. Leo builds a brick 5 cubes long, 5 cubes deep and 6 cubes tall. To make it strong, the red outside is **two** cubes thick everywhere (also at the bottom, top, front and back). Everything inside that is blue.



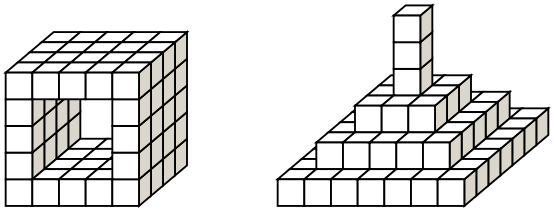
How many blue cubes are there?

A. 1 ***B.*** 2 ***C.*** 12 ***D.*** 36 ***E.*** 148

16. Grace paints a flag with three stripes: top, middle and bottom. Grace has 5 colours (white, red, black, green and yellow). All three stripes must be different colours.
How many different flags can Grace paint?

A. 125 ***B.*** 120 ***C.*** 74 ***D.*** 60 ***E.*** 15

17. Tom has the tunnel on the left. Tom takes it apart to build the pyramid on the right, but there are not enough cubes. Nothing is hollow inside the pyramid.



tunnel pyramid

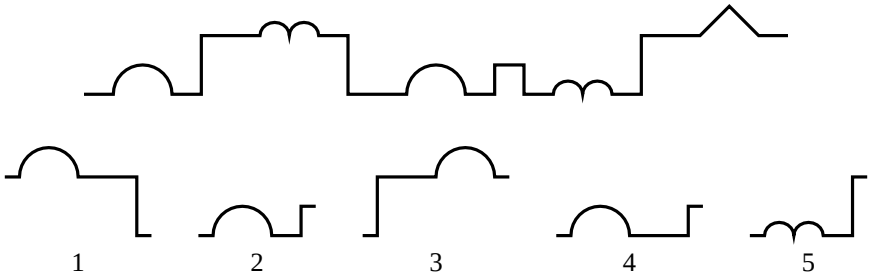
How many more cubes does Tom need?

A. 70 ***B.*** 38 ***C.*** 22 ***D.*** 20 ***E.*** 6

18. Lily has the cards 1, 2, 3 and 6. Using every card exactly once, with +, –, ×, ÷ and brackets, Lily wants to get the biggest possible number.
What is the biggest number Lily can get?

A. 12 ***B.*** 35 ***C.*** 37 ***D.*** 54 ***E.*** 62

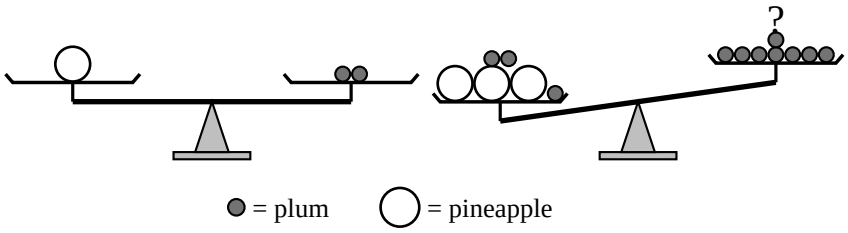
19. Lena cut a strip of paper along this line.



How many of the pieces 1–5 below belong to the line?

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

20. The first balance is level. On the second balance there are 3 pineapples and 3 plums on the left and 8 plums on the right.



How many plums must be added to the right plate to make it level?

A. 17 ***B.*** 9 ***C.*** 3 ***D.*** 2 ***E.*** 1

Answer Key

1.	**D**	**2.**	**E**	**3.**	**B**	**4.**	**A**	**5.**	**A**	**6.**	**B**
7.	**E**	**8.**	**C**	**9.**	**D**	**10.**	**B**	**11.**	**B**	**12.**	**A**
13.	**B**	**14.**	**D**	**15.**	**B**	**16.**	**D**	**17.**	**C**	**18.**	**D**
19.	**B**	**20.**	**E**								

Solutions

1. D (6 m 93 cm) — 50 dm = 5 m, 30 mm = 3 cm, so the training jump was 5 m 63 cm. Adding 130 cm gives 6 m 93 cm. [2003 Q10 · jump]

2. E (8 marbles) — Take 1 ball off both plates: 8 marbles balance 2 – 1 = 1 ball. So one ball = 8 marbles. [2002 Q9 · one_more_balances]

3. B (36) — (1 + 8) × 4 = 36. None of the other numbers can be made. [2002 Q21 · which_can]

4. A (Chloe) — The only possible matching: Nora – December; Sofia – January; Chloe – March; Julien – August; Aisha – July. [2002 Q18 · who_has]

5. A (24) — 4 children can stand first, then 3 are left for the second place, and so on: 4 × 3 × 2 × 1 = 24. [2002 Q22 · photo_line]

6. B (4 m 77 cm) — 30 dm = 3 m, 50 mm = 5 cm, so the training jump was 3 m 55 cm. Adding 122 cm gives 4 m 77 cm. [2003 Q10 · jump]

7. E (523) — The numbers with all digits the same are 8888, 9999, ... The next one after 9476 is 9999: 9999 – 9476 = 523. [2002 Q24 · same_digits]

8. C (48 cm) — The two new edges (2 cm and 2 cm) replace the two removed pieces of the outer edge, which were also 2 cm and 2 cm. So the perimeter is the same as the rectangle's: 2 × (18 + 6) = 48 cm. [2002 Q6 · corner_cut]

9. D (4) — A drawing is wrong if part of the pole edge is bare or if it reaches more than 4 squares from the pole. Wrong: B. So 4 can be right. [2002 Q15 · count_possible]

10. B — Only this piece is found in the line. The others don't fit — A: it has a pointed roof where the line has two small bumps; C: it has two small bumps where the line has a round dome; D: its flat part between the two features has the wrong length; E: its step is taller than in the line. [2002 Q7 · can_belong]

11. B (3) — The blue inside is 3 long and 2 tall, so one layer of it has 3 × 2 = 6 cubes. 6 ÷ 6 = 1 layers of blue, plus the red front and back: 3. [2003 Q9 · how_deep]

12. A (2512) — The numbers with all digits the same are 11111, 22222, ... The next one after 19710 is 22222: 22222 – 19710 = 2512. [2002 Q24 · same_digits]

13. B (2) — Each half must have 17. The ways: 8+7+2 | 2+6+4+5; 7+2+2+6 | 4+5+8. [2002 Q11 · count_ways]

14. D (Noah was born in March) — The only possible matching: Emma – April; Omar – March; Nicolas – December; Noah – June; Fabienne – October. [2002 Q18 · which_not_true]

15. B (2) — Two layers come off at each end of every length, so each length drops by 4: 1 × 1 × 2 = 2. [2003 Q9 · thick_wall]

16. D (60) — Top stripe: 5 choices, middle: 4, bottom: 3. 5 × 4 × 3 = 60. [2002 Q22 · stripes]

17. C (22) — Tunnel: 16 × 4 = 64 cubes. Pyramid: 7 × 7 + 5 × 5 + 3 × 3 + 1 × 1 × 3 = 86. Missing: 86 – 64 = 22. [2002 Q20 · not_enough]

18. D (54) — (1 + 2) × 3 × 6 = 54. Multiplying by 1 changes nothing, so it is better to add the 1 to the smallest other card first. [2002 Q21 · biggest]

19. B (2) — Pieces 2, 5 fit. Not: 1 (it has the right shapes but in the wrong order); 3 (it steps up where the line steps down); 4 (its flat part between the two features has the wrong length). [2002 Q7 · how_many_fit]

20. E (1) — One pineapple = 2 plums, so the left plate is worth 3 × 2 + 3 = 9 plums. The right plate has 8, so add 9 – 8 = 1. [2002 Q9 · what_to_add]